

# Carryover Half-Life and the Power of the Biomarker-Treatment Interaction Test: Closed-Form Results for the Hybrid Design

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## 1. Summary

This paper isolates problem 2 of docs/36, the collapse of power under small carryover reported by Hendrickson et al. (2020) for the Hybrid (N-of-1) design, and studies it analytically. The setting is reduced to the smallest case that still shows the problem: the Hybrid design only,  $N = 70$ , the published compound-symmetry (CS) correlation construct, and the paired-difference interaction statistic (E9) of the compendium. For that statistic every moment that matters has a closed form. The results:

1. **The half-life enters in one place only.** The variance of each patient's on-minus-off contrast  $\Delta$  does not depend on the carryover half-life  $t_{1/2}$  at all: carryover acts on the response means, which the contrast's intercept absorbs, and on the biomarker coupling. The slope of  $\Delta$  on the biomarker is

$$\gamma = -c_{bm} \frac{\sigma_{BR}}{\sigma_B} (1 - \bar{g}_{\text{off}}),$$

where  $\bar{g}_{\text{off}}$  is the mean biomarker coupling over the patient's off-drug visits: the average fraction of the biomarker effect present in the biomarker's correlation with the drug response when the patient is off drug (defined in Section 2; derived in Section 4). Power is a strictly decreasing function of  $\bar{g}_{\text{off}}$ , so the entire effect of carryover on power runs through this one number (Section 5).

2. **Under the published coupling rule, power does not decline with the half-life; it switches off.** The published rule sets the coupling to its full value wherever any residual drug effect remains, so  $\bar{g}_{\text{off}} = 1$  for every  $t_{1/2} > 0$ , however small. The slope is then exactly zero, and power equals the test size, for every positive half-life (Section 5.1). The published "precipitous decline" is a step at  $t_{1/2} = 0$ , not a decline. The residual power in the simulated published analysis (0.08-0.11 at  $c_{bm} = 0.25$ ) comes from the published analysis counting the baseline visit as off drug, which leaves a slope of  $1/(n_{\text{off}} + 1)$  of its no-carryover value; the closed form reproduces the published mean estimate to within 0.003.
3. **Under graded coupling, power declines monotonically, and the decline is provable.** With the coupling decaying as the drug effect does,  $\bar{g}_{\text{off}}$  increases strictly with  $t_{1/2}$ , so the slope shrinks, the residual variance

grows and power falls, strictly, for every  $t_{1/2}$  (Section 5.2). The loss is negligible at the half-lives Hendrickson examined (power 0.721 at  $t_{1/2} = 0.1$ , the same as without carryover, and 0.712 at 0.2) and substantial only when carryover is long (0.473 at one week, 0.278 at two). The published step rule is the  $t_{1/2} \rightarrow \infty$  limit of graded coupling: the published construct behaves, whenever carryover is nonzero, as if it were infinitely long.

4. **Under mean moderation, power does not depend on the half-life** (Section 5.3).
5. **The closed form reproduces the simulations.** It agrees with a direct Monte Carlo of the statistic, and its baseline-inclusive variant reproduces the published linear mixed-model (`lmer`) results for the Hybrid design: mean estimate, its standard deviation and power (Section 6).

## 2. Setting

**Design.** The Hybrid design of Hendrickson et al. (2020): eight post-baseline visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, with expectancy 1 at the two open-label visits and 0.5 afterwards, and four paths.  $N = 70$  patients are allocated 18, 18, 17, 17 across the paths, as in the published code. The time since discontinuation  $t_{sd}$  at each visit, from the published `buildtrialdesign()` semantics (verified):

| Path | On drug<br>(visits 1-8) | $t_{sd}$ (weeks)   | $n_{\text{on}}$ | $n_{\text{off}}$ |
|------|-------------------------|--------------------|-----------------|------------------|
| 1    | 1 1 1 1 0 0 1<br>0      | 0 0 0 0 1 2 0<br>4 | 5               | 3                |
| 2    | 1 1 1 1 0 0 0<br>1      | 0 0 0 0 1 2 6<br>0 | 5               | 3                |
| 3    | 1 1 1 0 0 0 1<br>0      | 0 0 0 1 2 3 0<br>4 | 4               | 4                |
| 4    | 1 1 1 0 0 0 0<br>1      | 0 0 0 1 2 3 7<br>0 | 4               | 4                |

Every off-drug visit follows an exposure, so every off-drug visit has  $t_{sd} \geq 1$  week. This matters below.

**Data-generating process.** The published construct (commit 58b32a9, configuration A of docs/36): patient-level baseline severity BL, biomarker  $B$ , and three response factors  $TV_t$ ,  $PB_t$ ,  $BR_t$  at each visit, jointly normal; outcome  $Y_t = \text{BL} - (TV_t + PB_t + BR_t)$ , and  $Y_0 = \text{BL}$  at baseline. Standard deviations  $\sigma_{TV} = 10$ ,  $\sigma_{PB,t} = 10e_t$ ,  $\sigma_{BR} = 8$ ,  $\sigma_B = 15.36$ . Correlations: within a factor,  $\rho = 0.8$  between any two visits (compound symmetry); between factors,  $c_1 = 0.2$  at the same visit and  $c_\times = 0.1$  at different visits. BL is uncorrelated with everything else. The biomarker is correlated with  $BR_t$  only,

$$\text{Cor}(B, BR_t) = c_{bm} g_t,$$

**Coupling.** We call  $g_t$  the *coupling* at visit  $t$ : the fraction of the biomarker effect  $c_{bm}$  that is present in the biomarker’s correlation with the drug response at that visit, with  $0 \leq g_t \leq 1$ . The term is ours, not Hendrickson et al.’s; `docs/36` writes the same quantity as the coupling vector  $r$ , with  $r_t = c_{bm}g_t$ . In the covariance construct the biomarker does not enter the outcome equation; it moderates the drug effect only through this correlation, so a patient with a high biomarker tends to have a larger drug response at the visits where the coupling is present. The interaction the analysis is meant to detect is the difference between on-drug and off-drug coupling: if the biomarker predicts the response as strongly off drug as on drug, there is no biomarker-by-treatment interaction to find. All three rules below set  $g_t = 1$  on drug; they differ in the off-drug coupling, and that difference is the subject of this paper. Under mean moderation the interaction is not carried by a correlation, and  $g_t = 0$  throughout.

The coupling pattern  $g_t$  is set by one of three rules:

- **Step (published).**  $g_t = 1[\mu_{BR,t} \neq 0]$ : full coupling wherever the mean drug response is nonzero, which off drug is the case whenever  $t_{1/2} > 0$ .
- **Graded.**  $g_t = 1$  on drug and  $g_t = \phi_t = 2^{-t_{sd}/t_{1/2}}$  off drug: coupling decays with the drug effect (adjustment 2 of `docs/36`).
- **Mean moderation.**  $g_t = 0$ ; instead the on-drug  $BR_t$  is shifted by  $c_{bm}\sigma_{BR}z$ ,  $z$  the standardized biomarker (adjustment 3 of `docs/36`).

For example, path 1’s off-drug visits fall 1, 2 and 4 weeks after discontinuation. At  $t_{1/2} = 0.5$  weeks their coupling is 1, 1 and 1 under the step rule, and 0.25, 0.06 and 0.004 under the graded rule.

Throughout,  $c_{bm} = 0.25$ . Under CS the joint distribution exists only for  $c_{bm}$  below a ceiling, 0.256 in the Hybrid design without carryover (`docs/36`, Section 6.5); 0.25 is the largest round value feasible for all three rules at every half-life, so no matrix is repaired anywhere in this paper. The published values 0.3 and 0.6 are infeasible here, and the published results at those values come from repaired matrices whose coupling the formulas below would have to take from the repaired matrix.

### 3. The statistic

The compendium’s paired-difference specification E9 (`analysis/report/compendium-mathematics.tex`, Section “Analysis specifications for carryover”) is a two-stage estimator. For patient  $i$ ,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it}, \quad a_t = \begin{cases} 1/n_{\text{on}} & \text{on drug} \\ -1/n_{\text{off}} & \text{off drug} \end{cases},$$

followed by the ordinary least squares (OLS) regression  $\Delta_i = \beta_0 + \beta_1 B_i + \varepsilon_i$  over all  $N$  patients and a  $t$ -test of  $\beta_1$  with  $N - 2$  degrees of freedom.

The slope  $\beta_1$  is the biomarker-by-treatment interaction: the change in a patient's on-minus-off outcome difference per unit of biomarker, in outcome points per biomarker unit. Since  $Y = \text{BL} - (\dots)$  is a symptom score that falls as patients improve, a biomarker that predicts a larger drug response gives a negative slope. Its expected value is written  $\gamma$  below; without carryover it equals  $-c_{bm}\sigma_{BR}/\sigma_B$ , the drug-response gain in standard-deviation units per standard deviation of biomarker, rescaled to the outcome and biomarker scales. It is the same quantity the published mixed model estimates as the `bm:Db` coefficient.

Two variants are treated:

- **E9**, the specification as written, on the eight post-baseline visits.
- **E9b**, the same with the baseline visit counted as one more off-drug observation ( $n_{\text{off}} + 1$  off-drug rows). This is how the published analysis codes it: `lme_analysis()` adds a baseline row with  $t = 0$  and  $D_b = 0$  before fitting  $Sx \sim \text{bm} + \text{Db} + \text{t} + \text{bm}*\text{Db} + (1|\text{ptID})$ . E9b is not that mixed model, but Section 6 shows that its moments and power match the mixed model's closely in this setting, so it serves as a closed-form proxy for the published analysis.

In both variants the weights over all rows sum to zero, so BL, which enters every row including baseline, cancels:

$$\Delta = - \sum_{t=1}^8 a_t S_t, \quad S_t = TV_t + PB_t + BR_t.$$

For E9,  $\sum_{t \geq 1} a_t = 0$ . For E9b the post-baseline weights sum to  $1/(n_{\text{off}} + 1)$ , the baseline row carrying the remaining  $-1/(n_{\text{off}} + 1)$  against an outcome of BL alone.

## 4. Closed-form moments

$\Delta$  and  $B$  are jointly normal within a path, so the regression of  $\Delta$  on  $B$  is exactly linear,

$$E(\Delta | B) = \mu + \gamma(B - \mu_B), \quad \text{Var}(\Delta | B) = \tau^2,$$

and three quantities per path determine everything: the variance of  $\Delta$ , its covariance with  $B$ , and its mean.

#### 4.1 The variance of the contrast

With  $A_1 = \sum a_t$ ,  $A_2 = \sum a_t^2$ ,  $A_e = \sum a_t e_t$ ,  $A_{ee} = \sum a_t^2 e_t^2$  and  $A_{2e} = \sum a_t^2 e_t$  (sums over the eight post-baseline visits), expanding  $\text{Var}(a^\top S)$  over the CS structure gives

$$\begin{aligned} \text{Var}(\Delta) = & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho A_1^2 + (1 - \rho)A_2] + \sigma_{PB}^2[\rho A_e^2 + (1 - \rho)A_{ee}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_\times A_1^2 + (c_1 - c_\times)A_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_\times A_1 A_e + (c_1 - c_\times)A_{2e}]. \end{aligned}$$

For E9,  $A_1 = 0$ , and the person-level part of every factor (the  $\rho$  terms) cancels except where the placebo factor's visit-dependent scale  $e_t$  keeps  $A_e$  away from zero. Evaluated at the published values (verified, by hand for path 1 and by the script for all paths):

| Variant | Paths 1, 2 | Paths 3, 4 |
|---------|------------|------------|
| E9      | 44.13      | 45.03      |
| E9b     | 51.04      | 51.77      |

**Proposition 1.** *Var( $\Delta$ ) does not depend on  $t_{1/2}$ .* The formula contains the response standard deviations and correlations only; the half-life enters the published construct through the response means ( $\mu_{BR,t}$ , with its recursive carry-over) and through the coupling  $g_t$ , and neither appears. The script confirms the value is identical at every half-life to the last digit (verified). The argument is not specific to CS: it holds for any construct in which  $t_{1/2}$  does not enter the response block, which includes every configuration of `docs/36`.

Counting the baseline as off drug (E9b) increases the variance by about 15%, because the baseline row adds BL-free weight that no longer cancels the person-level parts ( $A_1 \neq 0$ ).

#### 4.2 The slope

Only  $BR_t$  is correlated with  $B$ , with covariance  $c_{bm}g_t\sigma_B\sigma_{BR}$ , so  $\text{Cov}(\Delta, B) = -c_{bm}\sigma_B\sigma_{BR}\sum_t a_t g_t$  and

$$\gamma = -c_{bm} \frac{\sigma_{BR}}{\sigma_B} \sum_{t=1}^8 a_t g_t.$$

On-drug visits have  $g_t = 1$  and weights summing to 1. Writing  $\bar{g}_{\text{off}}$  for the mean coupling over the  $n_{\text{off}}$  post-baseline off-drug visits, and recalling that the baseline has  $g_0 = 0$ :

**Proposition 2.**

$$\gamma_{E9} = -c_{bm} \frac{\sigma_{BR}}{\sigma_B} (1 - \bar{g}_{\text{off}}), \quad \gamma_{E9b} = -c_{bm} \frac{\sigma_{BR}}{\sigma_B} \left(1 - \frac{n_{\text{off}}}{n_{\text{off}} + 1} \bar{g}_{\text{off}}\right).$$

Without carryover  $\bar{g}_{\text{off}} = 0$  and both equal the expected interaction slope  $-c_{bm}\sigma_{BR}/\sigma_B = -0.1302$ . Under mean moderation the coupling is zero and the on-drug shift contributes  $-c_{bm}\sigma_{BR}/\sigma_B$  directly, so  $\gamma = -0.1302$  whatever the half-life.

### 4.3 The residual variance and the correlation

For the covariance rules (step and graded),  $B$  explains part of the existing variance of  $\Delta$ , so

$$\tau^2 = \text{Var}(\Delta) - \gamma^2 \sigma_B^2.$$

Under mean moderation the shift adds variance rather than explaining it, so  $\text{Var}(\Delta)$  grows by  $\gamma^2 \sigma_B^2$  and  $\tau^2$  is the covariance-rule  $\text{Var}(\Delta)$  itself. This is why, at the same slope, mean moderation has slightly less power than the covariance rules without carryover (Section 5).

Power depends on these only through the correlation between the contrast and the biomarker,

$$\rho_{\Delta B} = \frac{\gamma \sigma_B}{\sqrt{\text{Var}(\Delta)}} = -c_{bm} \frac{\sigma_{BR}(1 - \bar{g}_{\text{off}})}{\text{SD}(\Delta)} \quad (\text{E9, covariance rules}),$$

0.30 in every path without carryover.

### 4.4 The pooled slope and its power

For a single path of  $n$  patients, write  $S_B = \sum_i (B_i - \bar{B})^2$ . Conditional on the biomarker values the OLS slope is exactly normal,  $\hat{\beta}_1 | B \sim N(\gamma, \tau^2/S_B)$ , and its  $t$ -statistic is exactly noncentral  $t$  with  $n - 2$  degrees of freedom and noncentrality  $\gamma\sqrt{S_B}/\tau$ . Since  $S_B \sim \sigma_B^2 \chi_{n-1}^2$ , power is the noncentral- $t$  tail probability averaged over that  $\chi^2$  law, and  $\text{Var}(\hat{\beta}_1) = \tau^2 E(1/S_B) = \tau^2 / \{\sigma_B^2(n-3)\}$ . At  $S_B = \sigma_B^2(n-1)$  the noncentrality is

$$\delta = \sqrt{n-1} \frac{\rho_{\Delta B}}{\sqrt{1 - \rho_{\Delta B}^2}},$$

so power is governed by the contrast-biomarker correlation alone.

E9 pools the four paths with a common intercept. With  $\pi_p = n_p/N$  and  $\mu_p = E(\Delta)$  in path  $p$ , the pooled slope has mean  $\bar{\gamma} = \sum_p \pi_p \gamma_p$ , and the differences between paths act as additional residual variance,

$$\sigma_{\text{res}}^2 = \sum_p \pi_p [\tau_p^2 + (\mu_p - \bar{\mu})^2 + \sigma_B^2 (\gamma_p - \bar{\gamma})^2],$$

so that the  $t$ -statistic is, to first order in the path differences, noncentral  $t$  with  $N - 2$  degrees of freedom and noncentrality  $\bar{\gamma} \sqrt{S_B} / \sigma_{\text{res}}$ ,  $S_B \sim \sigma_B^2 \chi_{N-1}^2$ , and

$$\text{Var}(\hat{\beta}_1) \approx \frac{\sum_p \pi_p [\tau_p^2 + (\mu_p - \bar{\mu})^2]}{(N - 3) \sigma_B^2} + \frac{2}{N} \sum_p \pi_p (\gamma_p - \bar{\gamma})^2.$$

Power is computed by averaging the noncentral- $t$  tail probability over 400 quantiles of  $\chi_{N-1}^2$ . A first version of the calculation used  $N$  in place of  $N - 3$  and ignored the variation of  $S_B$ ; it overstated power by about 0.01-0.02, which the direct simulation of Section 6 detected. The path-mean term is small: at most 0.30 against residual variances of 41-49, under 0.7% of the total, and it decreases with  $t_{1/2}$  (verified).

## 5. The effect of the half-life

By Propositions 1 and 2,  $t_{1/2}$  affects power only through  $\bar{g}_{\text{off}}$ , and  $|\rho_{\Delta B}|$ , the noncentrality and power are all strictly decreasing in  $\bar{g}_{\text{off}}$ . The three coupling rules differ only in how  $\bar{g}_{\text{off}}$  depends on  $t_{1/2}$ .

Closed-form power and mean slope (verified):

| Rule                | Variant | $t_{1/2} =$ |       |       |       |       |       |
|---------------------|---------|-------------|-------|-------|-------|-------|-------|
|                     |         | 0           | 0.1   | 0.2   | 0.5   | 1     | 2     |
| Step<br>(published) | E9      | 0.721       | 0.050 | 0.050 | 0.050 | 0.050 | 0.050 |
|                     | E9b     | 0.654       | 0.081 | 0.081 | 0.081 | 0.081 | 0.081 |
| Graded              | E9      | 0.721       | 0.721 | 0.712 | 0.631 | 0.473 | 0.278 |
|                     | E9b     | 0.654       | 0.653 | 0.647 | 0.586 | 0.473 | 0.328 |
| Mean<br>moderation  | E9      | 0.681       | 0.681 | 0.682 | 0.682 | 0.682 | 0.682 |
|                     | E9b     | 0.619       | 0.619 | 0.620 | 0.620 | 0.621 | 0.621 |

| Rule                     | Variant | Mean<br>slope,<br>$t_{1/2} =$ |        |        |        |        |        |
|--------------------------|---------|-------------------------------|--------|--------|--------|--------|--------|
|                          |         | 0                             | 0.1    | 0.2    | 0.5    | 1      | 2      |
| Step<br>(pub-<br>lished) | E9      | -0.130                        | 0      | 0      | 0      | 0      | 0      |
|                          | E9b     | -0.130                        | -0.029 | -0.029 | -0.029 | -0.029 | -0.029 |
| Graded                   | E9      | -0.130                        | -0.130 | -0.129 | -0.118 | -0.098 | -0.072 |
|                          | E9b     | -0.130                        | -0.130 | -0.129 | -0.121 | -0.106 | -0.085 |
| Mean<br>moder-<br>ation  | both    | -0.130                        | -0.130 | -0.130 | -0.130 | -0.130 | -0.130 |

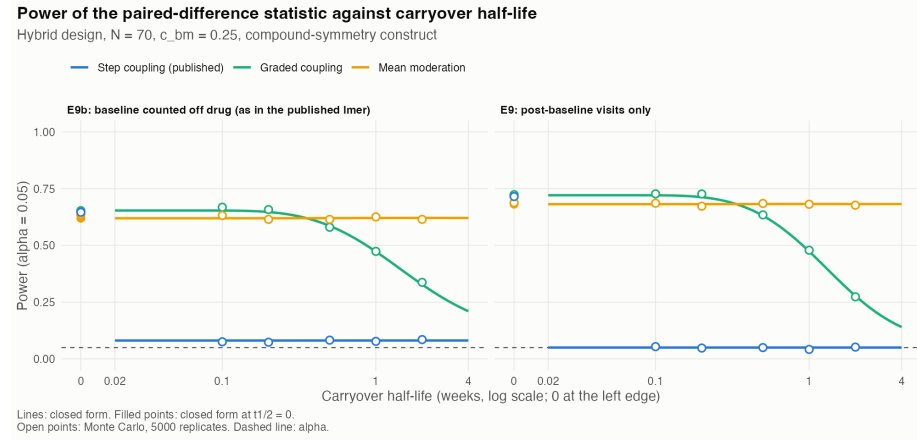


Figure 1: Power of the paired-difference statistic against carryover half-life

Figure 1. Closed-form power (lines; filled points at  $t_{1/2} = 0$ ) against carryover half-life on a logarithmic axis, with direct Monte Carlo estimates (open points). Hybrid design,  $N = 70$ ,  $c_{bm} = 0.25$ , CS construct.

### 5.1 Step coupling: a switch, not a decline

Under the published rule  $g_t = \mathbf{1}[\mu_{BR,t} \neq 0]$ . With  $t_{1/2} > 0$  the off-drug mean is  $\mu_{BR,t} = \mu_{BR,t-1} 2^{-t_{sd}/t_{1/2}} > 0$  at every off-drug visit, because every off-drug visit in the Hybrid design follows an exposure. Hence  $\bar{g}_{\text{off}} = 1$  for every  $t_{1/2} > 0$ , and by Proposition 2

$$\gamma_{E9} = 0, \quad \gamma_{E9b} = -c_{bm} \frac{\sigma_{BR}}{\sigma_B} \cdot \frac{1}{n_{\text{off}} + 1} \quad \text{for all } t_{1/2} > 0.$$

Power is therefore a step function of the half-life: its no-carryover value at  $t_{1/2} = 0$ , and a constant at every positive half-life, whether 0.01 weeks or 10. For E9 that constant is the nominal size, 0.050, because the slope is exactly zero; for E9b it is 0.081, carried by the baseline row alone, which is the only observation whose coupling stays at zero. The magnitude of the half-life plays no role. What the published figure shows as sensitivity to small carryover is the coupling rule redefining the estimand the moment any carryover is present: with constant coupling the biomarker moderates on-drug and off-drug response equally, and there is no interaction left to detect.

One numerical footnote. Below about 0.012 weeks the recursive off-drug mean at the latest off-drug visits underflows to exactly zero in double precision (it is about  $2^{-1300}$  at the last off-drug visit of path 4 when  $t_{1/2} = 0.01$ ), so the step rule, as coded both here and in the published code, switches those visits' coupling off. Figure 1 therefore starts at 0.02 weeks. This is a floating-point effect, not a property of the model.

The residual published power is explained exactly. E9b predicts mean slopes of  $-0.1302/4 = -0.0325$  in paths 1 and 2 and  $-0.1302/5 = -0.0260$  in paths 3 and 4, pooled  $-0.029$ ; the published analysis, simulated with the published construct, gives  $-0.026$  to  $-0.032$  (Section 6).

## 5.2 Graded coupling: a provable monotone decline

Under graded coupling,

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t_{\text{off}}} 2^{-t_{sd,t}/t_{1/2}}, \quad \frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}} t_{1/2}^2} \sum_{t_{\text{off}}} t_{sd,t} 2^{-t_{sd,t}/t_{1/2}} > 0,$$

since every off-drug visit has  $t_{sd,t} \geq 1$ .

**Proposition 3.** *Under graded coupling, in every path,  $|\gamma_p|$  is strictly decreasing and  $\tau_p^2$  strictly increasing in  $t_{1/2}$ ; hence the noncentrality, and the power of the path-stratified test, are strictly decreasing in  $t_{1/2}$  on  $(0, \infty)$ . With  $\bar{g}_{\text{off}}$  strictly increasing, Proposition 2 gives the slope, Section 4.3 gives  $\tau_p^2 = \text{Var}(\Delta) - \gamma_p^2 \sigma_B^2$  with  $\text{Var}(\Delta)$  fixed (Proposition 1), and the noncentral  $t$  tail probability is increasing in  $|\delta|$ . For the pooled E9 statistic as specified, with its common intercept, the path-mean term of Section 4.4 also moves with  $t_{1/2}$ ; it is under 0.7% of the variance and decreases with  $t_{1/2}$ , and power is non-increasing at every step of a 120-point grid from 0.02 to 4 weeks for both E9 and E9b (verified), with steps too small to resolve only at the shortest half-lives, where the loss is  $O(2^{-1/t_{1/2}})$ .*

The limits make the relation to the published rule precise:

- $t_{1/2} \rightarrow 0^+$ .  $\bar{g}_{\text{off}} \leq 2^{-1/t_{1/2}}$ , so the loss vanishes faster than any power of  $t_{1/2}$ : at  $t_{1/2} = 0.1$  week,  $\bar{g}_{\text{off}} \leq 2^{-10} \approx 0.001$  and power is unchanged to within 0.001. Power is continuous at zero.
- $t_{1/2} \rightarrow \infty$ .  $\bar{g}_{\text{off}} \rightarrow 1$ , so graded coupling converges to the step rule. The published construct is the infinite-half-life limit of graded coupling, applied at every positive half-life.

Two features are worth noting. The decline is slow at first and faster later because  $\bar{g}_{\text{off}}$  is a sum of  $2^{-t_{sd}/t_{1/2}}$  terms, each negligible until  $t_{1/2}$  is a substantial fraction of  $t_{sd}$ , which is at least one week here. And E9 and E9b cross: E9 has more power while carryover is short, because its variance is lower; E9b has more beyond about one week, because its baseline row keeps zero coupling however long the carryover and so retains part of the slope.

### 5.3 Mean moderation: no dependence

Under mean moderation the interaction is a shift of the on-drug mean only, so  $\gamma = -c_{bm}\sigma_{BR}/\sigma_B$  and  $\tau^2$  are both free of  $t_{1/2}$ , and power is constant (0.681-0.682 for E9). The recursive carryover of the mean response off drug changes  $\mu_p$  but not  $\gamma_p$  or  $\tau_p^2$ ; it enters only through the negligible path-mean term.

### 5.4 Power as a function of the coupling gap

Sections 5.1 to 5.3 are three special cases of one relation. With on-drug coupling 1, the slope is proportional to the *coupling gap*  $\bar{g}_{\text{on}} - \bar{g}_{\text{off}}$ , and power depends on the coupling profile only through it: neither which off-drug visits are coupled nor how the coupling is spread across them matters under E9, which weights every off-drug visit equally. Holding the response structure fixed and setting the off-drug coupling to  $1 - \text{gap}$  at every off-drug visit gives power as a function of the gap alone (closed form, verified):

| Coupling gap | 0     | 0.25  | 0.5   | 0.75  | 1     |
|--------------|-------|-------|-------|-------|-------|
| Power, E9    | 0.050 | 0.094 | 0.236 | 0.468 | 0.721 |
| Power, E9b   | 0.081 | 0.161 | 0.294 | 0.468 | 0.654 |

The coupling rules are points on this curve. Without carryover the gap is 1. Under graded coupling it is 0.991 at  $t_{1/2} = 0.2$ , 0.906 at 0.5, 0.754 at 1 and 0.551 at 2 weeks (patient-weighted over paths). Under the published step rule it is 0 at every positive half-life.

*Figure 2. Closed-form power against the coupling gap, the mean on-drug minus mean off-drug coupling over the post-baseline visits, with the positions of the coupling rules marked (E9). Under E9b the baseline row keeps zero coupling,*

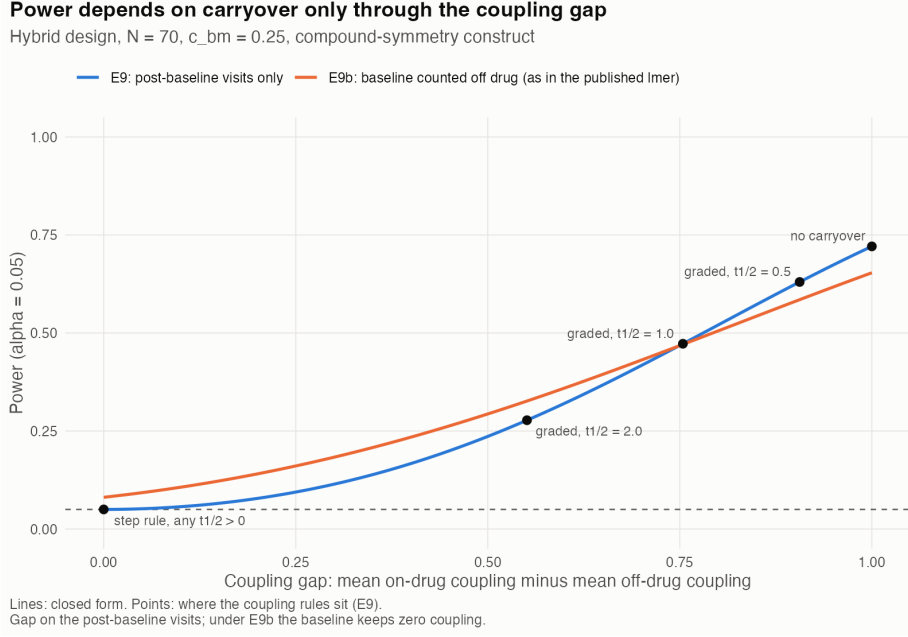


Figure 2: Power against the coupling gap

so its power does not fall to the test size at gap 0. Hybrid design,  $N = 70$ ,  $c_{bm} = 0.25$ .

Three things follow. First, power falls steeply with the gap: the noncentrality is proportional to it, so the sample size needed for a given power scales roughly as  $1/\text{gap}^2$ , and halving the gap calls for about four times as many patients. Second, the curve is steepest near gap 1 and flattest near 0: the first 0.1 of gap lost costs 0.097 of power (0.721 to 0.624), the last 0.1 only 0.007 (0.057 to 0.050). Short carryover nevertheless costs almost nothing under graded coupling, because the gap itself barely moves (0.991 at  $t_{1/2} = 0.2$ ), not because the curve is flat there. Third, the published step rule sits at the bottom of the curve for every positive half-life. The difference between the published result and a smooth decline is the difference between jumping from gap 1 to gap 0 and moving gradually along the curve. The curve is computed for the covariance construct; under mean moderation the gap is always 1 and the residual variance slightly larger (Section 4.3).

## 6. Validation

**Direct Monte Carlo.** The statistic was simulated directly from the published construct for each rule at  $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1, 2\}$ .

Each cell used 5000 simulated trials of 70 patients, drawn from the full joint distribution (not from the formulas), giving Monte Carlo standard errors of about 0.007 on power, 0.0007 on the mean slope and 0.0005 on its SD. E9 (verified):

| Rule            | $t_{1/2}$ | Mean,<br>closed | Mean,<br>MC | SD,<br>closed | SD,<br>MC | Power,<br>closed | Power,<br>MC |
|-----------------|-----------|-----------------|-------------|---------------|-----------|------------------|--------------|
| Step            | 0         | -0.1302         | -0.1299     | 0.0507        | 0.0508    | 0.721            | 0.715        |
| Step            | 0.1       | 0               | 0.0010      | 0.0531        | 0.0536    | 0.050            | 0.054        |
| Step            | 0.5       | 0               | -0.0010     | 0.0531        | 0.0525    | 0.050            | 0.050        |
| Step            | 2         | 0               | -0.0008     | 0.0531        | 0.0533    | 0.050            | 0.052        |
| Graded          | 0         | -0.1302         | -0.1301     | 0.0507        | 0.0498    | 0.721            | 0.723        |
| Graded          | 0.2       | -0.1290         | -0.1296     | 0.0508        | 0.0496    | 0.712            | 0.726        |
| Graded          | 0.5       | -0.1180         | -0.1190     | 0.0511        | 0.0516    | 0.631            | 0.634        |
| Graded          | 1         | -0.0982         | -0.0977     | 0.0517        | 0.0510    | 0.473            | 0.479        |
| Graded          | 2         | -0.0717         | -0.0724     | 0.0524        | 0.0516    | 0.278            | 0.273        |
| Mean            | 0         | -0.1302         | -0.1307     | 0.0531        | 0.0521    | 0.681            | 0.688        |
| moder-<br>ation |           |                 |             |               |           |                  |              |
| Mean            | 1         | -0.1302         | -0.1314     | 0.0531        | 0.0538    | 0.682            | 0.681        |
| moder-<br>ation |           |                 |             |               |           |                  |              |
| Mean            | 2         | -0.1302         | -0.1284     | 0.0531        | 0.0529    | 0.682            | 0.677        |
| moder-<br>ation |           |                 |             |               |           |                  |              |

Over all 36 cells (three rules, six half-lives, both variants; full table in `e9-closed-form-vs-mc.csv`), the mean slopes agree to within 0.002, the SDs to within 0.002, and power to within 2.7 Monte Carlo standard errors, with deviations of both signs averaging about +0.1 standard error. Four of the 36 power cells lie beyond 2 standard errors and one at exactly 2.0, against about two expected if the cells were independent; they are not (E9 and E9b share data), but a residual error of order 0.01 in the power approximation cannot be ruled out. The qualitative results of Section 5, the exact zero slope under the step rule, the floor at the test size and the monotone decline under graded coupling, are reproduced without exception.

**Against the published mixed-model analysis.** `docs/36` simulated the published construct (configuration A, step rule) and mean moderation (configuration D) with the published `lmer` analysis, 500 replicates per cell, at  $c_{bm} = 0.25$ . E9b reproduces it (verified):

| Rule            | $t_{1/2}$ | <code>lmer</code><br>mean | E9b<br>mean | <code>lmer</code><br>SD | E9b<br>SD | <code>lmer</code><br>power | E9b<br>power |
|-----------------|-----------|---------------------------|-------------|-------------------------|-----------|----------------------------|--------------|
| Step            | 0         | -0.126                    | -0.130      | 0.055                   | 0.055     | 0.632                      | 0.654        |
| Step            | 0.1       | -0.028                    | -0.029      | 0.055                   | 0.057     | 0.080                      | 0.081        |
| Step            | 0.2       | -0.026                    | -0.029      | 0.058                   | 0.057     | 0.096                      | 0.081        |
| Step            | 0.5       | -0.029                    | -0.029      | 0.058                   | 0.057     | 0.108                      | 0.081        |
| Step            | 1         | -0.032                    | -0.029      | 0.059                   | 0.057     | 0.108                      | 0.081        |
| Mean            | 0         | -0.131                    | -0.130      | 0.059                   | 0.057     | 0.628                      | 0.619        |
| moder-<br>ation |           |                           |             |                         |           |                            |              |
| Mean            | 0.1       | -0.132                    | -0.130      | 0.058                   | 0.057     | 0.672                      | 0.619        |
| moder-<br>ation |           |                           |             |                         |           |                            |              |
| Mean            | 0.2       | -0.127                    | -0.130      | 0.058                   | 0.057     | 0.636                      | 0.620        |
| moder-<br>ation |           |                           |             |                         |           |                            |              |
| Mean            | 0.5       | -0.131                    | -0.130      | 0.059                   | 0.057     | 0.646                      | 0.620        |
| moder-<br>ation |           |                           |             |                         |           |                            |              |
| Mean            | 1         | -0.129                    | -0.130      | 0.058                   | 0.057     | 0.652                      | 0.621        |
| moder-<br>ation |           |                           |             |                         |           |                            |              |

The Monte Carlo standard error of the `lmer` power is about 0.02 and of its mean estimate about 0.0026. The mean estimates agree to within 0.003 and the standard deviations to within 0.002. Power agrees to within 0.03 in nine of the ten cells; the exception is mean moderation at  $t_{1/2} = 0.1$  (0.672 against 0.619, about 2.6 Monte Carlo standard errors of the `lmer` estimate). Under mean moderation the mixed model is ahead in every cell (0.63-0.67 against 0.62), plausibly because it weights the visits by their covariance where E9b uses a simple on-off contrast; this was not tested separately. The structure of the published curve, a level of about 0.63 without carryover and a flat floor near 0.08-0.11 at every positive half-life, is exactly the structure Section 5.1 derives.

## 7. Discussion

**What problem 2 is.** In the simplest setting that shows it, the power collapse has a complete analytic account. Carryover does not degrade the information in the data: the variance of the contrast is fixed, and the half-life enters only through the biomarker coupling at off-drug visits. The published rule makes that coupling an indicator of whether any drug effect remains, so the interaction the simulation encodes vanishes at the first instant of carryover. The analysis then correctly finds nothing to detect. The problem is in the data-generating rule, not in the trial design or the analysis.

**What a decline should look like.** If the biomarker’s association with the drug response fades as the drug effect fades, power falls monotonically and smoothly with the half-life, by Proposition 3, starting from no loss at all for half-lives short relative to the off-drug intervals. At the published half-lives (0.1 and 0.2 weeks, against off-drug intervals of at least one week) the loss is under 0.01. This is the behavior the published discussion appears to have expected, and it is what the corrected coupling of `docs/36` produces in the full simulation.

**One number to watch.** Because power depends on carryover only through  $\bar{g}_{\text{off}}$ , a design can be judged by how much coupling it leaves at its off-drug visits: off-drug visits soon after discontinuation (small  $t_{sd}$ ) are the ones that lose their contrast first as carryover lengthens. In the Hybrid design these are the first blinded-discontinuation visits, one week after stopping.

**Covariance and mean moderation.** With the same slope, the covariance rules give slightly more power than mean moderation without carryover (0.721 against 0.681 for E9), because under covariance moderation the biomarker explains part of the variance the contrast already has, whereas under mean moderation it adds variance on top of it.

## 8. Limitations

- **CS only.** The construct is the published compound-symmetry structure. Proposition 1 holds for any construct whose response block does not depend on  $t_{1/2}$ , including the AR(1) and separable forms of `docs/36`, but the variance formula of Section 4.1 is specific to CS.
- **One design, one sample size, one effect size.** Hybrid,  $N = 70$ ,  $c_{bm} = 0.25$ . The structure of the results (the dependence on  $\bar{g}_{\text{off}}$ , the step and the monotone decline) does not depend on these choices; the numbers do.
- **Below the ceiling only.** The formulas assume the joint distribution exists. At the published  $c_{bm} = 0.3$  and  $0.6$  it does not; the published results there use repaired matrices, whose couplings differ from  $c_{bm}g_t$ .
- **E9 is not the published mixed model.** E9b approximates it and reproduces its moments in this setting, but the closed form is exact for the paired-difference statistic, not for `lmer`.
- **Large-sample pooling.** The pooled-slope variance is first order in  $1/N$  and power uses a normal approximation to the slope; the single-path results are exact.
- **Binary on/off contrast.** E9 contrasts on-drug against off-drug means. A contrast weighted by the expected exposure (the compendium’s  $D_{bc}$ ) would change the weights  $a_t$  and is not treated.

## 9. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-form.R \
--mc-reps 5000
```

Run from the repository root. The script reads the published construct from `analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R` (which reproduces the matrices of the 58b32a9 code exactly; docs/36, Section 6.5) and the `lmer` results from `analysis/data/quick-sim/hendrickson-problems/power-sim/power-summary.R`. It writes the per-path moments, the closed-form curves and both validation tables to `analysis/data/quick-sim/carryover-closed-form/` and Figure 1 to `docs/figures/37-fig1-power-closed-form.png`.

## 10. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. Code: [github.com/rchendrickson/pmsimstats](https://github.com/rchendrickson/pmsimstats), commit 58b32a9.
2. pmsimstats team. *Compendium mathematics*, `analysis/report/compendium-mathematics.tex` (specification E9; Paper 08's closed-form RM-ANOVA power result).
3. pmsimstats team. *Positive definiteness and carryover sensitivity in the Hendrickson et al. (2020) simulation framework*, `docs/36-hendrickson-pd-and-carryover.md`.